

Rapid Prototyping of Low-Fidelity Virtual Reality Hardware: Lessons Learned

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ABSTRACT

Rapid prototyping is an affordable and efficient method to explore novel designs for extended reality (XR) hardware and interactions. Prototyping is especially advantageous for smartphone-based virtual reality (VR) design as the hardware is inherently low-cost. The advancement of mobile VR hardware and interactions is an underrepresented area of research that has great potential to improve the inclusion of XR devices. Prototyping provides an opportunity for designers to explore potential solutions to mobile VR interaction and hardware in an efficient and cost-effective manner. In this work, we reflect on our experiences in prototyping a low-fidelity VR input device for mobile VR and identify a three-phase iterative process for prototyping low-fi VR input devices. We encourage members of the community to adopt this process to explore novel methods for low-fi VR hardware and interactions.

Keywords: Virtual reality, rapid prototyping, pointing device.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—virtual reality

1 INTRODUCTION

Smartphone-based virtual reality (VR), known as mobile VR (MVR), uses a smartphone placed inside a cardboard or plastic shell as the hardware and provides VR experiences at a low cost (\$10-55 USD). MVR has potential to democratize extended reality (XR) experiences, which has become an increasingly important issue as XR technologies are now being promoted for every-day work and use. However, the current limited state of interaction in low-fi VR prevents it from providing effective VR experiences comparable to current commercial VR head-mounted displays (HMD) [4, 9].

MVR can support 3 degrees-of-freedom (DOF) rotational head tracking and has one button on the HMD for binary or continuous input. Since it lacks 3D-tracked controllers provided with virtually all modern VR devices, MVR does not support the most common and simplest VR interaction techniques ray-casting and virtual hands. The selection of virtual objects or UI elements is typically performed using head-gaze pointing while the button or a dwell time is used for selection indication. MVR applications commonly lack any form of locomotion and are typically passive look-around experiences [4, 5, 9].

Most VR applications require interaction support that is not available in MVR. This prevents a large portion of the population who cannot afford modern VR HMDs from experiencing the proven benefits of immersive VR for collaborative work [3],

learning [1, 10, 11], health and therapy interventions [6, 8], and entertainment [7]. Individuals of lower socioeconomic status or living in remote regions cannot participate in what may be the next major computing paradigm for work, socializing, and everyday use.

More research into affordable solutions for improving MVR hardware, software, and interactions is needed to help democratize VR and XR experiences. Given the low-fidelity nature of MVR, this problem poses an opportunity to take advantage of rapid prototyping in the design of MVR hardware and interactions. The benefits of prototyping in the design process have long been known in the human-computer interaction (HCI) community, as it allows designers to quickly test design variations in a cost-effective manner. Designers can quickly explore the breadth of a design space, rapidly iterating on promising designs, and discarding less successful designs without committing significant time into their development. However, unlike hardware for high-end VR systems, low-fidelity prototypes of novel MVR hardware (e.g., input devices) can potentially simulate a user experience that is accurate to what the final product will provide (depending on software fidelity).

We previously designed and evaluated a low-fi controller for MVR [2]. During our iterative design process, we performed rapid prototyping to quickly test and compare design variations to find an optimal solution.

In the following sections, we describe our design process and experiences prototyping for low-fi VR interactions. We then propose a three-phase method for XR input device prototyping that can be widely adapted into low-fi VR design.

2 PROTOTYPING A LOW-FI VR CONTROLLER

We designed a 6DOF cardboard controller for MVR using optical marker tracking and evaluated the selection performance and user experience of the controller [2]. We had two main goals to achieve through our design process. First, was to determine an ideal marker configuration that could maintain tracking while the controller was moved and oriented in different directions. Second, designing an effective method for selection indication as, unlike modern VR controllers, the controller has no physical controls like a button to facilitate “clicking”.

The initial stage of our design process involved testing different tracking software to determine the best option for our controller. During this phase, we used very rudimentary prototypes to abstractly represent the controller as the focus was not yet on the hardware design. This involved printing paper markers and attaching them to small handheld household objects or even our own hands. Our intentions were never to use these objects as part of the controller’s design, but rather these make-shift controller representations allowed for quick and easy testing of tracking software options.

After choosing a tracking solution, we began testing different marker configurations. The initial design was very simplistic: a rectangular piece of cardboard and one paper marker attached to it. When testing this design, we determined more markers were

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required to achieve the tracking fidelity we desired. This led to generating and testing multiple controller prototypes of various sizes and shapes and with varying numbers of markers and marker sizes. Since the prototypes were all made of cut cardboard and printed paper, assembling each prototype was quick and required no additional cost. In addition, small adjustments were made to the prototypes in real time to test potential design improvements without the need to build a new prototype from scratch. For example, an additional cardboard piece was attached to the controller prototype's handle using tape to change its size. While this was not visually ideal, it achieved the purpose of our prototyping phase to efficiently test and compare design variations.

Once we settled on a marker configuration and controller shape, we investigated different methods for selection indication, effectively a process to "click" a button to complete selection. We developed three methods (Flip, Thumb-over, Trigger) for selection indication. All methods required an additional camera-tracked marker, either placed on the controller or on a second cardboard piece held in the nondominant hand. The marker must be hidden from the smartphone camera to confirm a selection. Once the user has identified the object to select (e.g., by pointing a ray or intersecting the virtual object and controller), selection indication occurs once the marker is occluded from camera view.

The Flip and Thumb-over selection indication methods were similar, the defining difference being a square cut piece of paper being taped to the user's thumb (see Figure 1). With Flip, the user turned the cardboard piece away from the camera to hide the marker and confirm a selection. In contrast, with Thumb-over, the user moved their thumb over the marker to hide it with the square paper piece. This simple difference exemplifies an advantage of low-fidelity prototyping as we were able to effectively evaluate the Thumb-over method without needing a fully formed design and high-fidelity prototype.



Figure 1: Cardboard apparatus used for selection indication methods Flip (left) and Thumb-over (right).

The third technique, Trigger, is an example of a higher-fidelity prototype. For Trigger, we assembled a cardboard gun using cut cardboard pieces and an elastic band to test selection indication via a trigger (see Figure 2). Upon pulling the trigger, a piece of cardboard would shift downward to occlude the marker, which was placed on the top surface of the controller, to indicate selection. While the gun prototype took more time to assemble, it acted as a base that could be iterated upon to quickly test minor design modifications.

In general, over our design process we designed and assembled multiple low-fidelity prototypes to determine an optimal low-fi

VR controller hardware design, tracking implementation, and selection indication method. The low-cost and DIY nature of the materials enabled us to make rapid small design iterations to the prototypes without hassle or extra cost.



Figure 2: Gun controller prototype using Trigger selection indication and cardboard components it is assembled out of.

3 DISCUSSION

Through our experience of rapid prototyping of a MVR input device, we have formulated a three-phase iterative process for prototyping such devices and made observations on the nature of prototyping and its potential benefits for low-fi XR design.

3.1 Phases of Prototyping for VR Input Devices

We identified three iterative phases of prototyping for VR input devices: the exploratory, refinement, and rapid adjustment phases.

3.1.1 Phase 1: Exploratory Phase

In the exploratory phase, rapid low-fidelity prototypes are created to test the feasibility of potential hardware designs. This is the most rudimentary phase of prototyping. The prototypes are primitive, disposable, and only serve to test ideas. This phase is iterative in nature and can be performed in parallel to an initial brain-storming phase regularly seen in HCI design.

During the exploratory phase of prototyping for our low-fi VR controller, we created make-shift prototypes to test different tracking software and marker configurations by sticking printed markers to objects and cutting unmeasured cardboard pieces. During this phase, our design space had the most breadth; we had not yet formed any concrete controller designs and were open to testing a wide variety of implementations.

3.1.2 Phase 2: Refinement Phase

After the design space has been comprehensively explored with rapid prototyping in the exploratory phase, the refinement phase is where a more formal prototype or prototypes are assembled and empirically evaluated. The design of the refined prototype is based on the findings of the exploratory phase and its assembly requires more time and effort than the primitive prototyping of phase 1. The prototypes made in phase 2 may involve precise measurements, cleaner designs, and be made of the materials intended for the final design.

In the refinement phase of prototyping our low-fi VR controller, we defined precise dimensions for the controller and marker sizes, and measured distances between each marker. More care was taken when cutting the cardboard pieces. Figure 3 shows the controller prototype in the right hand from this phase of our prototyping process.



Figure 3: Controller prototype using 3 cube markers (right) and apparatus for selection indication (left).

3.1.3 Phase 3: Rapid Adjustment Phase

Finally, the rapid adjustment phase is when small adjustments are made to the refined prototype from phase 2 to test potential design improvements to be implemented in future prototypes. The adjustments may involve adding, moving, or removing parts of the refined prototype to change its usability or user experience in some way without impeding its ability to be used and tested.

For example, when testing our refined controller prototype, the selection indication apparatus (see Figure 3) was originally a vertically-oriented rectangular shape with a single marker. However, while evaluating the usability of the prototype design, we determined adding a second marker placed closer to the centre of the field of view could improve the selection indication method. To test this hypothesis, we simply cut and attached an additional cardboard piece to change the apparatus to an L-shape. This provided space for the second marker. While the glued-together cardboard is less ideal than a single cardboard piece, this rapid adjustment allowed us to test a design variation without need to remake the prototype. If the small design variations prove beneficial, they can then be incorporated into an additional refined prototype or final design.

Another example of a rapid adjustment occurred while we were testing our gun controller prototype. Over extended use, the cardboard trigger began to deteriorate due to its thin shape. This indicated the thickness and shape of the trigger needed to be updated in a future prototype, but to continue testing the current prototype we wrapped it in tape to temporarily strengthen the trigger (see Figure 2). We also made minor adjustments to where the markers were attached to the top of the gun controller. This required moving and adding new cardboard pieces to the top of the controller, making the prototype visually less polished but remaining usable for further testing.

After rapid adjustments have been made and tested, the prototype process may loop back to phase 2 where an updated prototype is made that incorporates the minor adjustments from phase 3, or to phase 1 if the designers conclude a substantially different design is necessary. The three-phase iterative process is depicted in Figure 4.

3.2 Open Questions

While we have proposed three phases of prototyping that can be applied to low-fi VR hardware design, it is unclear if they would be as effective for the design of high-end VR devices. The low-fidelity nature of prototyping lends well to low-fi VR as similar materials are used for the prototypes and final design. However, this is far from true for high-end VR. We speculate our rapid prototyping process can potentially aid in the ergonomic design of input devices, but further exploration is needed to determine whether our proposed three-phase process is effective for the design of high-end VR input devices.

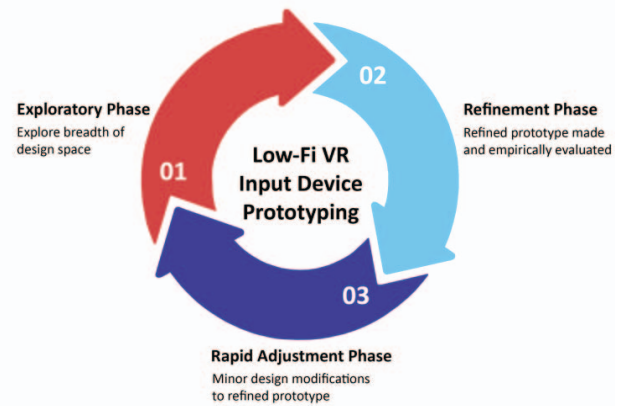


Figure 4: Three-phase iterative process for prototyping input devices for low-fi VR.

4 CONCLUSION

Prototyping in the design of XR input devices allows designers to rapidly explore the design space in a cost-effective manner that naturally melds into the iterative design approach commonly used in HCI design. The main advantage we found while prototyping was the ability to easily make adjustments to a prototype design and immediately test its efficacy. In general, prototyping is an affordable and efficient method to explore novel approaches to XR hardware and interaction.

Rapid prototyping perfectly lends to the low-cost, low-fi nature of MVR. The advancement of MVR interactions and application support is an underrepresented area of research in our community that has great potential to enhance inclusion of XR devices. We identified three iterative phases of prototyping for low-fi VR input devices that we encourage members of the community to adopt. This three-phase approach to prototyping can aid in the exploration and design of novel methods for low-fi VR hardware and interactions.

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